



OVERVIEW

DentOS uses an **offline, cryptographically-signed license** to unlock the app, plus an optional **cloud subscription** (Supabase) for sync and remote subscription control. This guide shows you, the vendor/operator, how to activate a build for your own testing now, and how to issue and manage paid subscriptions for clinics later.

Two identities — don't confuse them

A **license** unlocks the software on a device (works fully offline). A **cloud subscription** is a separate Supabase account per clinic that enables sync and lets you suspend/renew access remotely. Clinic **staff logins** are a third thing entirely — they live in the clinic's local database and never appear in Supabase auth.

A Activate a build for testing (now)

You run the signing tool once to create your vendor key pair, embed the public key in the app, then mint a test license and paste it into the activation screen.

1 Generate your key pair & sign a test license

From the project root, run the vendor tool (already in your repo at `tool/license_tool.dart`):

```
dart run tool/license_tool.dart
```

It prints a long **public modulus** and writes two files:

- `vendor_keypair.json` — your **private** signing key. Keep it secret; never commit it.
- `license.json` — the signed license you'll paste into the app.

2 Embed the public key in the app

Open `lib/licensing/data/license_verifier.dart` and paste the printed number into the modulus (keep the quotes):

```
static final BigInt _modulus =  
    BigInt.parse('PASTE_THE_PRINTED_MODULUS_HERE');
```

Stop and re-run the app after pasting (`flutter run`). The key is compiled in — a running app won't pick it up, and activation will report an invalid signature.

3 Activate

Launch the app. On the **activation screen**, open `license.json`, copy its entire contents, paste into the box, and press **Activate**. The wizard then continues to clinic profile and the owner account.

4 Create the owner login

Finish the wizard by setting the clinic name and the owner's username + password. That owner account is stored locally (encrypted) and is what you sign in with — fully offline. You now have a working build.

The modulus and license must come from the same run

If you delete `vendor_keypair.json` and regenerate, you get a new key — you must paste the new modulus *and* mint a fresh `license.json`. A mismatch is the #1 cause of “signature not valid”.

B Issue subscriptions to clinics (production)

For each paying clinic you do two things: **(1)** mint a signed license with their tier, seats, branches and expiry, and **(2)** provision their cloud account in Supabase. The **same clinic id** must be used everywhere.

1 Set the license fields, then sign

Edit the fields at the top of `tool/license_tool.dart` for that clinic, then run it to produce their `license.json`:

Field	What to set
<code>clinicId</code>	Unique id for the clinic, e.g. CL-0007. Used in the license, the cloud row, and the auth claim.
<code>clinicName</code>	Shown on invoices and the welcome screen.
<code>tier</code>	basic standard premium (drives which Premium features unlock).
<code>cloudPackage</code>	none or cloud (whether sync is included).
<code>maxBranches</code>	Branch limit the app enforces.
<code>maxUsers</code>	Seat limit the app enforces for staff logins.
<code>expiresAt</code>	Subscription end date — the app locks after this.

You keep one `vendor_keypair.json` for your whole business — every clinic license is signed by it, and the public modulus you already embedded verifies them all.

2 Provision the clinic in Supabase

In the Supabase dashboard, create the subscription row and the one device auth account:

```
-- 1) subscription record (the 48-hour gate reads this)
insert into clinics (id, name, tier, status, expires_at)
values ('CL-0007', 'Bright Smile Dental', 'premium', 'active',
        now() + interval '365 days');

-- 2) Authentication > Users > Add user (email + password)
-- then tag it with the clinic id:
update auth.users
set raw_app_meta_data = raw_app_meta_data || '{"clinic_id":"CL-0007"}'::jsonb
where email = 'clinic-device@yourdomain.com';
```

The Supabase **auth user** is the clinic’s device credential, not a staff login. There is exactly **one per clinic**. Without it, the app gets stuck on the connection check.

3 Deliver to the clinic

- Ship the build with their `.env` filled in: `SUPABASE_URL`, `SUPABASE_ANON_KEY`, and the device `SUPABASE_CLINIC_EMAIL` / `PASSWORD` you just created.
- Send them `license.json` to paste on the activation screen.
- They activate, set up their owner account, and start working. The owner can then create branch logins from Settings.

4 Renew or suspend later

- **Renew:** mint a new license with a later `expiresAt` and bump `expires_at` on the `clinics` row. The client pastes the new license on the **Locked** screen to instantly restore access — data is never touched.
- **Suspend (non-payment):** set `clinics.status = 'suspended'` (or let `expires_at` pass). The app locks within the 48-hour window; their data stays intact and returns the moment you reactivate.

C Tiers & the 48-hour rule

Tier limits are whatever you write into each license — the app enforces `maxBranches` and `maxUsers`. A typical layout:

Tier	Branches	Seats	Cloud sync
Basic	1	2	Optional
Standard	1	4	Optional
Premium	3+	8+	Included

Every tier needs internet at least every 48 hours

DentOS works offline, but must reach Supabase once inside any 48-hour window to re-verify the subscription. Past that, it shows the Reconnect screen until it gets online — by design, across all tiers.

D Security & troubleshooting

Protect the signing key — it is the root of trust for every license you issue.

- Keep `vendor_keypair.json` offline and backed up privately. Anyone with it can mint valid licenses.
- The public modulus is safe to ship — it only verifies, never signs.
- Add `vendor_keypair.json` and `.env` to `.gitignore`; never commit them.

Symptom	Most likely cause
"Signature not valid" on activate	Modulus in the app doesn't match the key that signed, or the app wasn't restarted after pasting the modulus.
"License has expired"	<code>expiresAt</code> is in the past — issue a renewal license.
Stuck on the connection check	Supabase auth user missing, <code>.env</code> not loaded, or (macOS) the network-client entitlement isn't set.
Activate does nothing	<code>license.json</code> wasn't pasted in full — copy the entire file contents.

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